

Encryption Scavenger Hunt

Objective

The purpose of this activity is to examine HTTPS certificates from three different websites and understand how Certificate Authorities (CAs), certificate validity periods, and the chain of trust are used to secure web communication.

Table

Website	Subject(Issued to)	Issuer(Issued by)	Valid from	Valid to
AWS	aws.amazon.com	Amazon RSA 2048 M04 Organization: Amazon	November 6, 2025	October 18, 2026
Prime Video	www.primevideo.com	Amazon RSA 2048 M04 Organization: Amazon	July 31, 2026	Feb. 14, 2027
HackTheBox	hackthebox.com	YE2 Organization: Let's Encrypt	Sep. 6, 2026	Dec. 5, 2026

Observations

The certificates demonstrate how HTTPS uses digital certificates to establish trust and secure communication. Each certificate is issued by a Certificate Authority (CA) and has a defined validity period.

I also observed that different websites can use different Certificate Authorities. AWS and Prime Video use certificates issued by Amazon, while Hack The Box uses a certificate issued by Let's Encrypt.

This is directly related to my MQTT security project. Instead of relying on a public CA, I will create my own private Certificate Authority and use it to issue a certificate for the Mosquitto MQTT broker. The MQTT clients will then be configured to trust this CA.

Connection to the MQTT Project

Real Website	My MQTT System
Certificate Authority	My private Certificate Authority
Website certificate	MQTT broker certificate
Browser trusts the CA	MQTT sensors/clients trust my CA
HTTPS/TLS	MQTT over TLS